src/flightCompV2/servos.h

```
/*
 * Header file for controling ARC-S Servo's
 *
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 */

#ifndef SERVOS_H
#define SERVOS_H

#include <ESP32Servo.h>
#include <EEPROM.h>

#define EEPROM_SIZE 64   // adjust if you add more data
#define EEPROM_START 0   // start address

struct AxisServo
{
  String name;
  Servo servo;
  int pin;
  int offset; // center in microseconds
  int direction;

  void init_servo()
  {
    servo.setPeriodHertz(50);
    servo.attach(pin);
    servo.writeMicroseconds(offset); // center on init
  }

  void writeDeltaUs(int deltaUs)
  {
    int pulse = offset + deltaUs;
    pulse = constrain(pulse, 500, 2500);
    servo.writeMicroseconds(pulse);
  }
};

// High-level commanded angles (deg) for each servo axis
struct ServoMix {
    float posX = 0.0f;
    float negX = 0.0f;
    float posY = 0.0f;
    float negY = 0.0f;
};
```

```
48
49 // Last commanded servo mix, kept for telemetry
50 extern ServoMix lastServoCommand;
51
52 //UTIL
53 inline int clampServoUs(int us);
54 void loadOffsetsFromEEPROM();
55 void readSerialAndUpdateServo();
56 void processCommand(String cmd);
57 void ControlServo(String name, float angle);
58 float mapGyroToServo(float angle, float minAngle, float maxAngle);
59 float clamp(float value, float minValue, float maxValue);
60
61
62 // CONTROL LOOP
63
64 const float KP_PITCH = 0.9f;
65 const float KP_YAW   = 0.9f;
66 const float KP_ROLL  = 0.5f;
67
68 const float KI_ROLL  = 0.0f;
69 const float KD_ROLL  = 0.02f;
70
71 const float SERVO_LIMIT_DEG = 60.0f;   // mechanical authority
72
73 void flightControlUpdate(double dt);
74
75 void pitchControlLoop(double dt);
76 void initPitchController();
77
78 void yawControlLoop(double dt);
79 void initYawController();
80
81 void rollControlLoop(double dt);
82 void initRollController();
83
84
85 void applyServoMix();
86
87 extern const int NUM_SERVOS;
88 extern AxisServo servos[];
89 #endif
```